

CHAPTER 40

Tinnitus retraining therapy

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Abstract: Tinnitus retraining therapy (TRT) is a specific clinical method based on the neurophysiological model of tinnitus described by Jastreboff (Jastreboff, P.J. (1990). *Neurosci. Res.*, 8: 221–254). The method is aimed at habituation of reactions evoked by tinnitus, and subsequently habituation of the tinnitus perception. Several other methods have been suggested for habituation of tinnitus, but in TRT two components that strictly follow the principles of the neurophysiological model of tinnitus are implemented and necessary: (1) counseling, aimed at reclassification of tinnitus to a category of a neutral signals and (2) sound therapy, aimed at weakening tinnitus-related neuronal activity as suggested by Jastreboff and Hazell (Jastreboff, P.J. and Hazell, J.W.P. (2004). Cambridge University Press, Cambridge). This chapter outlines the theoretical basis of TRT as well as comments on the clinical outcome of the use of TRT for different kinds of tinnitus.

Keywords: tinnitus retraining therapy; learning; conditioned reflexes; conscious and subconscious paths; the limbic and autonomic nervous systems

Introduction

The neurophysiological model of tinnitus

The main postulate of the neurophysiological model (Jastreboff, 1990) of tinnitus is that other systems in the brain, in addition to the auditory system, have to be involved in the clinically significant tinnitus, i.e., tinnitus which bothers people to such an extent that it interferes with their everyday life. Moreover, these systems, specifically the limbic and sympathetic part of the autonomic nervous systems, are actually responsible for negative, bothersome reactions to tinnitus (Jastreboff and Hazell, 2004).

The second postulate, that tinnitus is a phantom auditory perception (Jastreboff, 1990), is currently

accepted by a majority of investigators and clinicians (Muhlneckel et al., 1998; Möller, 2003, 2006; Eggermont and Roberts, 2004; Muhlau et al., 2006; Bartels et al., 2007; Weisz et al., 2007) and has many significant consequences. It points out that tinnitus perception results from the detection and perception of neuronal activity within the auditory pathways — similarly to perception of activity evoked by a sound, but in the case of tinnitus without being linked to any mechanical, vibratory activity present within the cochlea. This is in contrast to what has been labeled as “objective tinnitus” where the perception is caused by physical sounds that are generated in the body. “Objective tinnitus” is currently labeled a “somatosound” (Jastreboff and Jastreboff, 2003). The recognition that in case of tinnitus there is no vibratory activity within cochlea solved a number of tinnitus puzzles. For example, suppression of tinnitus perception reflects neuronal suppression and not

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auditory masking; therefore, there is no clear frequency relation between tinnitus pitch match and frequency of external sound and there is no critical band phenomenon (Feldmann, 1971). The phantom nature of the tinnitus perception explains why on many occasions it is impossible to suppress tinnitus perception even by very high levels of the sound, and why sound levels needed for suppression are higher than predicted from tinnitus loudness match. All these observations have significant implication on all treatments, which utilize sound to interfere with abnormal neural activity that causes the tinnitus, including masking or sound therapy of TRT.

Third postulate of the model differentiate between mechanisms involved in tinnitus perception and in tinnitus-induced suffering (e.g., annoyance, anxiety, problems with sleep and concentration). The mechanisms responsible for generation of tinnitus-related neuronal activity, perceived as tinnitus, are common for people who just experience tinnitus (~80% of all individuals with tinnitus) and those who suffer because of it. This is supported by a lack of differences in psychoacoustical characterization of tinnitus (i.e., its pitch and loudness match and minimal suppression “masking” level) between these two groups of individuals with tinnitus (Jastreboff et al., 1994). Specific mechanisms, which are responsible for spreading tinnitus-related neural activity to other structures in the brain, are present only in people who suffer from their tinnitus and are separate from mechanism involved in generation of the perception of abnormal neural activity that causes the tinnitus. Therefore, these two groups of mechanisms are discussed separately in this chapter.

Many mechanisms have been proposed as being responsible for the generation of abnormal neural activity that causes tinnitus (Kiang et al., 1970; Salvi and Ahroon, 1983; Møller, 1984, 1999; Tonndorf, 1987; Pujol, 1992; Jastreboff, 1995; Eggermont and Roberts, 2004; Eggermont, 2005, 2006; Heinz et al., 2005; Weisz et al., 2005; Ma et al., 2006; Zenner et al., 2006; Mazurek et al., 2006b; Sanchez et al., 2007). While the neurophysiological model of tinnitus does not depend on mechanism responsible for generating the neural activity that causes the perception of tinnitus, a

specific mechanism was proposed together with the model, called discordant damage (dysfunction) of outer and inner hair cell systems (Jastreboff, 1990). This hypothesis has been described previously in detail (Jastreboff and Hazell, 2004) and only an outline is presented here.

Discordant damage of outer and inner hair cells

Inner hair cells (IHC) are the auditory receptor cells that transduce the sound-evoked mechanical vibration in the cochlea into a neural code in auditory nerve fibers and thus subsequently make it possible to perceive sounds. Outer hair cells (OHC) on the other hand are mechanical amplifiers providing 50–60 dB of amplification. OHCs are helpful, but not necessary for hearing and have little effect on sounds louder than ~60 dB HL. OHCs are typically damaged first by ototraumatic events, but even with up to 30% damage to OHCs, the impact on the hearing threshold is minimal (Chen and Fechter, 2003).

The discordant dysfunction theory postulates that each area on the cochlear basilar membrane on which OHC are totally or partially dysfunctional and IHC are in better functional shape than OHC causes an imbalance of activity in the dorsal cochlear nucleus, specifically, disinhibition caused by decreased inhibitory input from OHC (Jastreboff and Hazell, 2004). This in turn may increase spontaneous activity (Kaltenbach, 2006) and result in emergence of bursting, epileptic-like activity, which is further processed within the auditory pathways (Chen and Jastreboff, 1995; Jastreboff, 2004). Currently, the available data support the “discordant dysfunction” theory, including its prediction that increase of the damage to the IHC system with constant level of damage of OHC could actually decrease tinnitus (Kaltenbach et al., 2002; Jastreboff and Hazell, 2004).

The discordant dysfunction theory can explain many observations, which are difficult to explain otherwise. For example, the presence of tinnitus in patients with normal hearing can be due to the presence of patches of dysfunctional OHC, the effect of which cannot be detected by standard audiometric tests (pure tone threshold) but can be

identified by otoacoustic emission testing. This can explain the association of distortion product otoacoustic emission (DPOAE) and tinnitus (Ozimek et al., 2006; Hall, personal communication). Indeed, high frequency resolution DPOAE performed on our patients since 1992 indicate higher extent of irregularity than in the DPOAE of individuals without tinnitus. On the other hand, the absence of tinnitus in 27% of people with total deafness (Hazell et al., 1995) can be explained noticing that the neural activity that causes the tinnitus depends on the difference between functionality of OHC and IHC. If both systems are totally destroyed then the abnormal neural activity that causes tinnitus would be less pronounced than what it would when only OHC are dysfunctional as there would be a smaller difference in activity coming from OHC and IHC systems. Consequently, this signal can easily undergo spontaneous habituation. In the majority of individuals with tinnitus, the pitch of the tinnitus tends to be in the frequency range of the greatest hearing loss. The biggest difference between OHC and IHC functionality is expected to occur in the region of the basilar membrane that is tuned to these frequencies. Typically, cochlear damage along the basilar membrane process through stages of: gradually increase of damage to OHC with little damage to IHC, and damage the IHC only occurs when all OHC are destroyed (Chen and Fechter, 2003). Thus, the frequency range of the greatest hearing loss, particularly the bottom of the slope of audiogram, corresponds to the area of the basilar membrane where the difference between damage of OHC and IHC systems is largest. Hearing loss from exposure to loud noise is an important factor associated with tinnitus (Hoffman and Reed, 2004). Noise exposure causes damage/dysfunction of OHC first and only at very high sound levels does the damage extend to IHC (Chen and Fechter, 2003). Therefore, according to discordant damage theory, noise causes the optimal situation to induce tinnitus. Furthermore, “disco tinnitus,” i.e., temporal tinnitus associated with exposure to loud music can be explained as loud sound causing disorganization of cilia of OHC. Under such situations, these cells are temporarily disabled and if they are not irreversible

damaged, the cilia return to normal state after few hours — days.

Salicylate (aspirin), quinine, and cisplatin are the most powerful drugs to induce tinnitus. Administration of salicylate in high doses always induce tinnitus (Mongan et al., 1973). These drugs, while working through different biochemical mechanisms (see Chapter 12), all mostly affect OHC with little or no effect on the function of IHC.

There are many other potential mechanisms of inducing abnormal neural activity that causes the tinnitus, including central tinnitus; however, it seems that discordant dysfunction theory is applicable to majority of individuals with tinnitus and it is useful in patient counseling. Note, that the validity of the neurophysiological model of tinnitus, and consequently TRT, do not depend on the proposed mechanisms generating the abnormal neural activity that causes the tinnitus.

Physiologic aspects of suffering from tinnitus

The hypothesis proposed in the neurophysiological model of tinnitus that the mechanisms of tinnitus-induced problems (e.g., annoyance, anxiety, panic, sleep, and concentration disturbances) involve other than auditory system emerged from analysis of clinical information. There is a difference between experiencing tinnitus and suffering from tinnitus. Tinnitus is bothersome for only ~20% of people experiencing tinnitus (Coles, 1996; Davis and El Refaie, 2000; Hoffman and Reed, 2004). There are no known differences in its pitch, matching of its loudness and the ability to mask the tinnitus with sounds of tinnitus that does not cause suffering and tinnitus that cause suffering. If auditory system was crucial in determining the extent of tinnitus-induced problems, the psychoacoustic of tinnitus should play a dominant role. It has also been observed that the severity of tinnitus and treatment outcome do not depend on the psychoacoustical characteristic of the tinnitus (Jastreboff et al., 1994). This observation argues against the auditory system being the anatomical location where aspects of tinnitus that leads to suffering are processed and interpreted. If the auditory nervous

system was the location where such aspects of tinnitus were processed it could be predicted that louder tinnitus would be more bothersome and more difficult to treat than softer tinnitus.

Finally, audiologic data show that average hearing is the same for the tinnitus and nontinnitus population (Jastreboff and Hazell, 2004). A comparison of the hearing thresholds of a population of people attending a tinnitus clinic, and who had bothersome tinnitus, with those from a gender and age matched group of individuals without tinnitus revealed that their hearing loss are essentially identical (Hazell and McKinney, 1996). While the prevalence of tinnitus perception increases with hearing loss, nevertheless occurrence of bothersome tinnitus does not seem to follow such simplistic rules (Hoffman and Reed, 2004).

The lack of relationship between psychoacoustic description of tinnitus and its severity and treatment outcome indicate that the auditory system does not play a primary role in tinnitus that cause suffering (clinically relevant tinnitus) but that non-auditory systems in the brain have a dominating role in creation of such qualities of tinnitus. The model postulated that tinnitus-induced problems (suffering) involved the limbic and autonomic nervous systems (Jastreboff, 1990, 1995). In particular, activation of the sympathetic part of the autonomic nervous systems is important in determining the severity of tinnitus. The limbic system controls emotional expression, memory storage and recall, motivation and mood, and it directly influences neuroendocrine and autonomic functions. The autonomic nervous system controls the action of glands, respiratory, circulatory, digestive, hormonal, and urogenital system (Møller 2003, 2006; Brodal 2004).

Development of tinnitus

Three stages in the development of tinnitus can be identified: (1) generation of the abnormal neural activity that causes the tinnitus; (2) interpretation of the abnormal neural activity; and (3) its perception and evaluation in high regions of the CNS. For clinically significant tinnitus (tinnitus that causes suffering) there is an additional stage (4)

sustained activation of brain regions that are not specifically auditory such as the limbic and autonomic nervous systems.

These steps are illustrated in Figs. 1A–D. Initially, the abnormal neural activity that causes tinnitus is typically generated at the periphery of the auditory system (Fig. 1A), perhaps in the dorsal cochlear nucleus (Jastreboff and Hazell, 2004; Kaltenbach, 2006). This signal may then be detected and further processed in the subconscious part of the brain (Chen and Jastreboff, 1995; Jastreboff, 2004) (Fig. 1B). Finally, it reaches the high cortical levels of the auditory system where it is perceived. Note, that in this idealistic situation other systems in the brain are not activated. Abnormal neural activity that causes the tinnitus is then treated as any other background sound and it is not evoking any reaction. However, under real-life conditions, when tinnitus is perceived as a new signal, it is evaluated, compared with information stored in memory, and attracts attention. Two potential scenarios are then possible. If the abnormal activity that causes tinnitus is classified by the conscious and the subconscious brain as a neutral stimulus, then it is subsequently blocked from reaching conscious perception (habituation of perception) and it is not spreading to other systems in the brain, as there is no need for any action in response to its presence. Specifically, the limbic and autonomic nervous systems are not activated by such neural activity (Fig. 1C). This scenario happens spontaneously in the majority of persons with tinnitus.

If however the abnormal neural activity that causes the tinnitus gets some negative connotation it is classified in the category of potentially unpleasant or even dangerous stimuli and consequently activates the limbic and autonomic nervous systems to assure readiness to reaction in response to its presence and attracts attention forcing its further evaluation (Fig. 1D) and a cascade of events occurs. If a person does not possess sufficient knowledge about the benign nature of tinnitus then, as with all unknown stimuli, the automatic assumption is that it might be something troublesome, attract attention and the process of conscious thinking about it and further analysis starts.

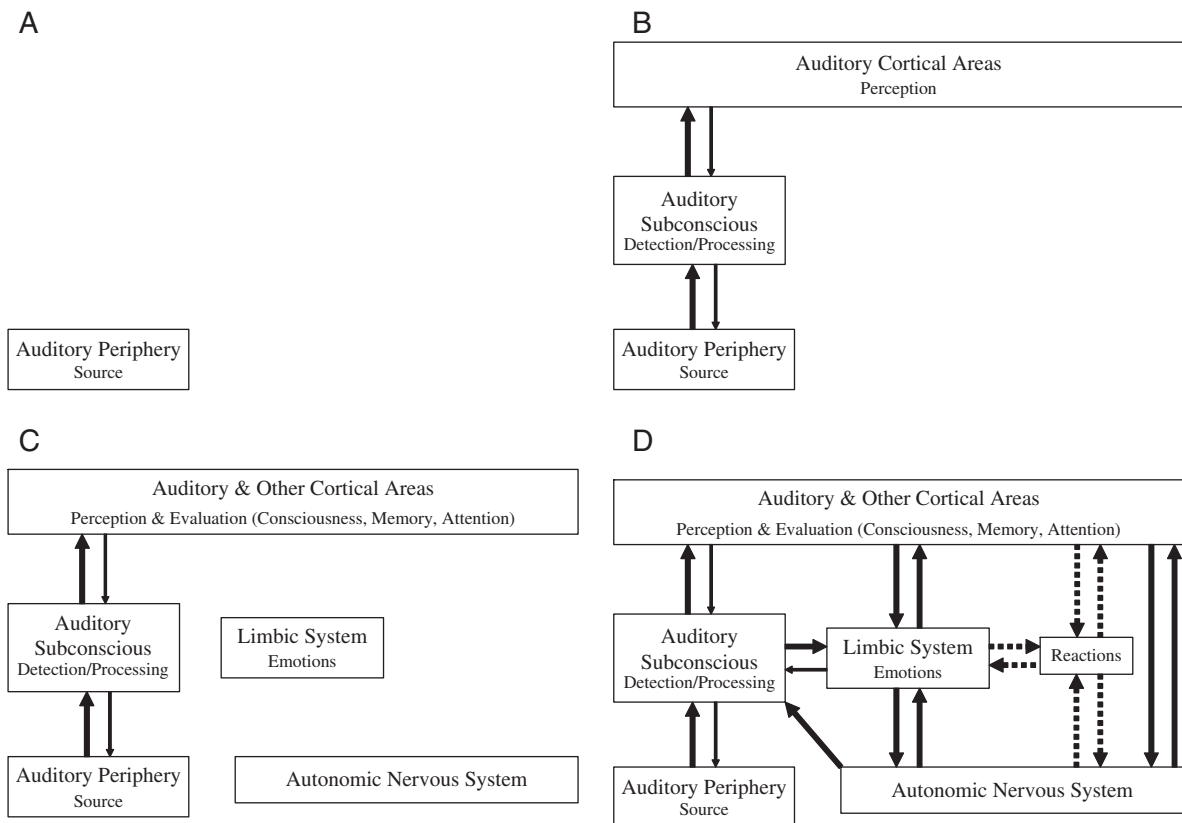


Fig. 1. Development of connections yielding tinnitus-induced negative reactions. (A) First stage, with tinnitus source being typically located in the periphery of the auditory system. (B) The abnormal neural activity that causes tinnitus is detected and processed at subconscious pathways of the auditory system and perceived at auditory cortex. (C) Evaluation of abnormal neural activity that causes the tinnitus as neutral stimulus prevents activation of the limbic and autonomic systems; this is situation in majority of tinnitus cases when spontaneous habituation of tinnitus occurs. (D) Classification of tinnitus as an important, negative stimulus yields development of self-enhancing loops governed by principles of the conditioned reflexes. Note, two loops — high, cortical involving consciousness, and subconscious loop.

Negative reactions induced by tinnitus depend on activation of limbic and the autonomic nervous system, and may be enhanced by prolonged activation of these systems. The fact that the functional connections between auditory system and limbic and autonomic structures are partly governed by the principles of conditioned reflexes (Chapters 1 and 4) a fact that has profound implications for the theoretical aspects of the proposed model as for clinical methods used for treatment of the kinds of tinnitus that cause suffering. A conditioned reflex that can evoke a reaction is created when there is a temporal coincidence of a sensory stimulus and reinforcement

(reward or punishment). The causal relation between stimulus and reinforcement is not necessary to create a reflex but temporal coincidence is sufficient. For example, perception of tinnitus while a person is under strong negative emotions would be sufficient to initiate a conditioned reflex. At the initial stage of tinnitus development, cognitive processes may play a dominant role including the fear of not being able to improve tinnitus and the potential medical problems linked to it.

Once the reflex is established then the sensory stimulus without the need for reinforcement is sufficient to evoke a negative reaction. Clinically significant tinnitus (suffering) is typically continuous

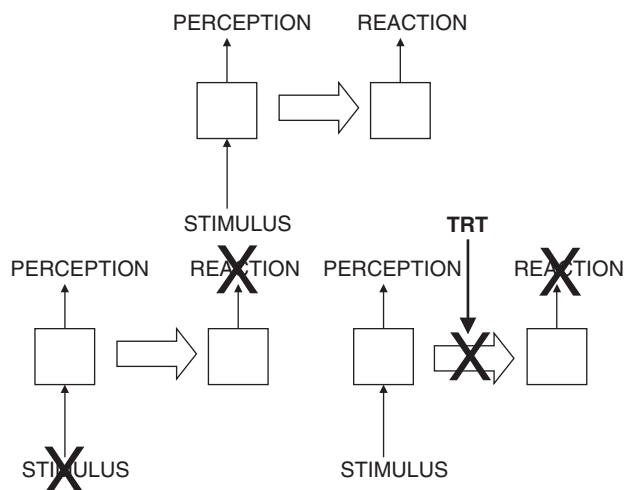


Fig. 2. The mechanisms of TRT action as compared with other approaches. Note, that TRT is neither aimed at modifying abnormal neural activity that causes the tinnitus nor at direct modification of tinnitus-evoked reactions but at blocking functional connections between sensory (auditory) and parts involved in generating reactions (the limbic and autonomic nervous systems).

and the reactions induced by the activation of the autonomic nervous system act as negative reinforcement. Tinnitus-related neuronal activity acts as a stimulus and the reaction evoked by activation of the sympathetic part of the autonomic nervous system act as reinforcement. Consequently, once the initial association between tinnitus perception and some negative event happening at the same time develops, the reflex loop is rapidly enhanced, as both stimulus and reinforcement are continuously present. Furthermore, the likelihood of spontaneous improvement of tinnitus, which would reflect the extinction of this reflex is low.

Most treatments proposed for tinnitus aim at reducing the tinnitus-related neuronal activity (e.g., by medications, electrical or magnetic stimulation and masking, see Chapters 34–39) or at decreasing tinnitus-induced reactions (e.g., psychological treatments aimed at improving coping, attention distraction, psychotropic medications acting at the limbic system). TRT differs from these approaches as it is aimed at changing specific functional connection between the auditory and the limbic and autonomic nervous systems without the attempt to change the abnormal neural activity that causes the tinnitus or directly attenuate tinnitus-evoked reactions (Fig. 2).

It has been recognized that processing of information at a subconscious level plays a significant role in learning, as well as in many other aspects of life. Reflex reactions and learning can occur without the conscious perception of a stimulus (Ohman, 1988; Esteves et al., 1994; Morris et al., 1998). Once the conscious and subconscious loops are created, the subconscious path could become dominant and therefore attenuating the conscious path alone might not be sufficient to alleviate the suffering from tinnitus. Results from Emory Tinnitus & Hyperacusis Center support this hypothesis by showing that a percentage of time during which the patients were aware of tinnitus and their subjective perception of its loudness did not have significant impact on the severity of the tinnitus as evaluated by the Tinnitus Handicap Inventory.

Physiological basis for TRT

The neurophysiological model of tinnitus described earlier (Jastreboff, 1990; Jastreboff and Hazell, 2004) has made it possible to develop a method of treatment for tinnitus. Currently, there is no reliable method for attenuating the tinnitus source and achieving a cure. It has been hypothesized that tinnitus as a problem arises because an

abnormal conditioned reflex arc is created. However, any kind of conditioned reflex can be reversed and retrained by proper techniques. The brain exhibits a high level of plasticity making it possible to habituate to any sensory signal, if this signal does not have negative implications (Chapter 2). Therefore, by interfering with the tinnitus-related neuronal activity that occurs above the tinnitus source, it should be possible to block the spreading tinnitus-related neuronal activity to the limbic and autonomic nervous systems (habituation of reactions) and prevent activation of high cortical areas where it would be perceived (habituation of perception) (Fig. 3).

Habituation is a normal and essential feature of the brain and occurs automatically in response to any neutral or low importance stimuli. Necessity of habituation results from the fact that we can perform only one task at a time that requires full attention. The problem is how to manage the enormous amounts of sensory input that is received all the time. The solution to this problem is to select important stimuli and block unimportant stimuli at the subconscious level so that it does not reach higher levels of the CNS and reaching our awareness inducing perception and inducing reactions, thus habituation occurs to unimportant stimuli. The important question is then what stimuli are regarded as unimportant.

TRT is aimed at inducing and facilitating habituation of tinnitus (both reactions induced

by tinnitus and its perception). The primary goal is to habituate reactions. Once this is achieved and abnormal neural activity that causes the tinnitus become more abnormal, the habituation of perception will follow automatically. Therefore, TRT utilizes the natural feature of the brain aiming at its utilization at abnormal neural activity that causes the tinnitus. There are two main components of TRT, both strictly based at the neurophysiological model of tinnitus: (1) counseling and (2) sound therapy. The goal of counseling is to reclassify tinnitus into category of neutral stimuli. As long as tinnitus is judged as important or potentially threatening, its habituation is difficult. The role of sound therapy is to decrease the strength of abnormal neural activity that causes the tinnitus in systematic manner over the period of the treatment. Sound therapy is used as well to treat hyperacusis, which accompanies tinnitus in ~30% of cases (Jastreboff and Jastreboff, 2002).

As the treatment is aimed to work above the source of tinnitus, the etiology of tinnitus is irrelevant and TRT can be successfully used for any type of tinnitus as well as for somatosounds (it should not be ignored that a somatosound may indicate severe but treatable disorders of the vascular system). The final clinical goal of TRT is to reach a stage when tinnitus does not interfere with the patient's life. Specifically, tinnitus, when present, should not cause annoyance and the patient's awareness of the tinnitus should be so low that it does not influence normal life. The results of treatment of 303 consecutive patients at Emory who had initial THI score at least 36 showed that significant improvement was achieved after 1 month of the treatment with THI score decreasing from 65 to 46, followed by further consistent improvement when followed up to 18 months hyperacusis was dramatically improved after 12 months, and in majority of cases a cure was noted. Both mean the change of THI score and the average change for all these patients reached level of statistical significance after 3 months. After 12 months, 82% of patients showed statistically a significant decrease of 20 points from the initial score.

Results of open studies reported from other centers also consistently showed significant improvement in over 80% of the patients who were treated

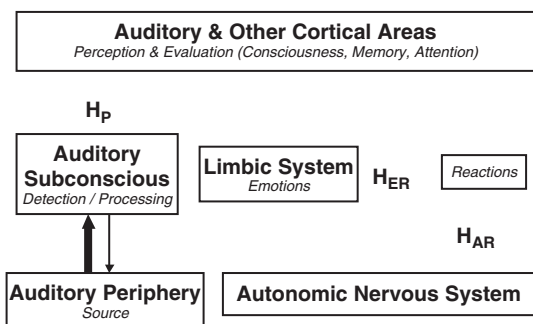


Fig. 3. Specific functional connections at which habituation of tinnitus occurs. H_p : habituation of tinnitus perception; H_{ER} : habituation of emotional reactions; H_{AR} : habituation of reactions evoked by the autonomic nervous system; H_p : habituation of tinnitus perception. Primary goal of TRT is to achieve habituation of reactions and then habituation of perception will follow automatically.

by TRT (Bartnik et al., 1999; Heitzmann et al., 1999; McKinney et al., 1999; Sheldrake et al., 1999; Herraiz et al., 2005; Mazurek et al., 2006a). The results of 5 years of follow up showed that the improvement is long lasting (Lux-Wellenhof and Hellweg, 2002). A systematic randomized study showed TRT to be highly effective, with highly statistically significant decline of THI and of percentage of time when tinnitus was annoying. Specifically, over period of 18 months in group with severe tinnitus THI decreased from 72 to 26.4; in group with mild tinnitus TRI decreased from 30.2 to 18.8. The percentage of time when patients were annoyed by tinnitus decreased from 47.3% to 6.3%. (Henry et al., 2006).

Conclusion

Tinnitus remains a challenging subject to study and to treat. The mechanisms are still poorly understood and there is no consensus regarding its optimal treatment. However, it appears that TRT provides an effective approach to alleviating the impact of tinnitus on patients' lives (the suffering) in a significant way. Additionally, TRT is also effective in treating hyperacusis. For TRT to be successful, it is important to follow the guidelines of the neurophysiological model of tinnitus for both counseling and sound therapy.

Abbreviations

CNS	central nervous system
DPOAE	distortion product otoacoustic emission
HL	hearing level
IHC	inner hair cells
OHC	outer hair cells
THI	tinnitus handicap inventory
TRT	tinnitus retraining therapy

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